Dongwook **Kwon**

AI Engineer · Agentic AI Systems

11F, 37, Geumto-ro 80beon-gil, Sujeong-gu, Seongnam-si, Gyeonggi-do, 13453, South Korea

(+82) 10-6612-9246 | dwkwon@suresofttech.com | dongwooky.github.io | dongwooky | dongwooky

Summary

Highly accomplished AI Research Engineer specializing in Large Language Model (LLM) optimization and Multi-Agent Systems (MAS), with a proven track record of bridging advanced academic research and high-impact industrial applications. Demonstrated research excellence through publications in top-tier natural language processing venues, including an Oral presentation at EMNLP 2025 and a peer-reviewed paper at ACL 2026 focusing on cost-aware LLM cascades. Inventor of multiple registered patents in AI-driven anomaly detection and a key technical lead who has successfully delivered mission-critical autonomous agent systems for global enterprise leaders, including Hyundai Motor Company and LG Electronics. Poised to significantly advance the United States' technological edge in critical and emerging fields by engineering scalable, reliable, and secure autonomous AI architectures.

Patents

Apr 2026 **Method for Implantable Medical Device to Detect Anomaly in Real Time**, KR 10-2961984 *Republic of Korea*

Aug 2025 **GPT API-Based Network Anomaly Detection**, KR 10-2848814 *Republic of Korea*

Publications

Journal Articles

- Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.
- **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

Conference Papers

- R. Chang†, **D. Kwon**†, J. Lee†, and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026.
  †Equal contribution.
- J. Lee†, R. Chang†, **D. Kwon**†, H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025.
  †Equal contribution.
- **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. ***Best Poster Award — Silver.***
- **D. Kwon**, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. ***Outstanding Paper Award.***

Experience

Suresoft Technologies *Seongnam-si, Korea*

AI Engineer, Intelligent Agent Technology Team *Dec. 2025 - Present*

- Developing an AI validation system utilizing AI agent-based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University *Seoul, Korea*

Research Assistant *Aug. 2021 - Dec. 2025*

- Researched deep-learning algorithms for anomaly detection in time-series data (ECG, cyber-physical systems).
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.
- Published in EMNLP 2025 (Oral) and ACL 2026; co-author on two granted Republic of Korea patents.

LG Electronics Canada (with University of Toronto) *Toronto, ON, Canada*

Visiting Researcher *Jan. 2025 - Jul. 2025*

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

Qualcomm Institute, University of California San Diego *San Diego, CA, USA*
AI DEVELOPMENT PROJECT PARTICIPANT, SUMMER PROGRAM *Jul. 2022 - Aug. 2022*

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Education

Kwangwoon University *Seoul, Korea*
MASTER OF SCIENCE IN COMPUTER ENGINEERING *Mar. 2024 - Feb. 2026*

- GPA: 4.33/4.5 (98.1%)

University of Toronto *Toronto, ON, Canada*
GRADUATE RESEARCH PROGRAM IN MECHANICAL AND INDUSTRIAL ENGINEERING *Jan. 2025 - Jul. 2025*

- GPA: 3.68/4.0

Kwangwoon University *Seoul, Korea*
BACHELOR OF SCIENCE IN ELECTRONICS AND COMMUNICATIONS ENGINEERING *Mar. 2019 - Feb. 2024*

- GPA: 3.41/4.5 (88.1%)

Honors & Awards

MAJOR AWARDS

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego *United States*
Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries *South Korea*

ACADEMIC RECOGNITION

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers *South Korea*
Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024) *South Korea*
Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering *South Korea*

COMPETITIONS

Dec. 2025 **Silver Award**, 23rd Embedded Software Competition (KIISE) *South Korea*
Dec. 2022 **Bronze Award**, Hanium ICT Mentoring Competition — FKII *South Korea*
Sep. 2022 **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX) — Kwangwoon University *South Korea*
Oct. 2021 **Grand Prize**, MY (Multi-Y) Capstone Design Competition — Kwangwoon University *South Korea*

Funded Research Projects

LG Electronics Canada *Toronto, ON, Canada*
AGENTIC AI EVALUATION FRAMEWORK *Jan. 2025 - Present*

- Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

Sejong University *Seoul, Korea*
AI-BASED INTRUSION DETECTION AND ATTACK PACKET CRAFTING TECHNOLOGIES FOR APR1400 *Jul. 2022 - Present*

- Developed an AI defense system to protect nuclear power plants from cyber attacks.

Ministry of Science and ICT *Seoul, Korea*
QUADRUPED ROBOT USING VISION SLAM WITH A DEPTH CAMERA *Apr. 2022 - Nov. 2022*

- Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Qualcomm Institute, University of California San Diego *San Diego, CA, USA*
PERSONALITY-BASED DRUG ADDICTION PREDICTION SYSTEM *Jul. 2022 - Aug. 2022*

- Developed a personality-based drug addiction prediction system using a machine learning model.

Kookmin University *Seoul, Korea*
DEVELOPMENT OF ACTIVE KILL-SWITCH AND BIOMARKER-BASED DEFENSE SYSTEM FOR LIFE-THREATENING IOT MEDICAL DEVICES *Sep. 2021 - Dec. 2022*

- Developed a security system for heart implant devices using an anomaly detection algorithm.

Ministry of Oceans and Fisheries *Seoul, Korea*

Development of a Smart Logistics System Using Computer Vision *Apr. 2021 - Nov. 2021*

- Developed a logistics detection system incorporating location tracking and stack-layer identification for containers.

Kwangwoon University *Seoul, Korea*

Smart Kitchen IoT System Utilizing Computer Vision and Deep Learning *May 2021 - Oct. 2021*

- Developed a cloud IoT platform featuring a system for classifying groceries in the fridge and an OCR receipt program.

Leadership Experience

Kwangwoon International Student Association

President

- Lead intercultural programs and student engagement initiatives.

Kwangwoon Broadcasting Center (KWBC)

Director of Filming and Editing

- Oversaw media production and content delivery.

Dept. of Electronics and Communications Engineering

Class Representative

- Represented student interests in academic affairs.

Amateur Radio Club

Executive Member

- Organized radio workshops and technical exhibitions.